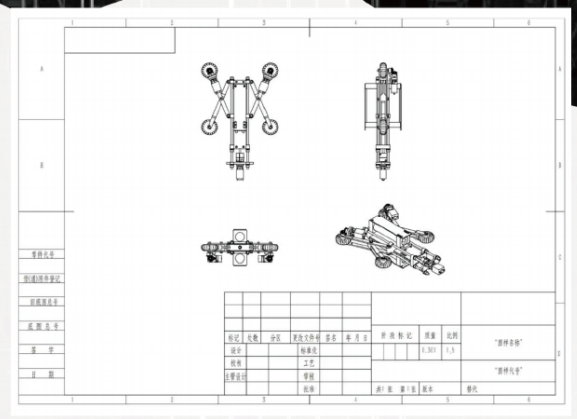
Internal Pipe Climber

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Project Overview:

This poster presents the design and technical details of a pipe-crawling robot. The robot is designed to move through pipes with the help of a unique locomotion system that integrates multiple essential components for stability and functionality. Below, we display the three-view drawing, key code snippets, essential components, and the BOM (Bill of Materials) for clarity and understanding.



Three-View Drawing of the Robot:

The poster includes a clear A3-sized three-view diagram of the robot. The front, top, and side views of the robot are displayed in an orderly and detailed manner, showing the key components and assembly layout.
The diagram includes a detailed technical drawing, showcasing the critical engineering features and parts.

Key Code Snippets:

The poster includes the essential code used to control the robot's movement, including logic for detecting black lines on the ground to guide the robot and control its motion.
The code includes basic movement functions like forward, backward, and stop, along with a detection routine for line following.
The primary logic of the code enables smooth control over the robot's movement, ensuring it follows predefined paths, like a black line.

```
#define SAMPLE_TIMES 5  // 采样次数
#define THRESHOLD 4     // 有效阈值

void forward(void)//前进
{
    BA1 = 1;
    BB1 = 0;
    BA2 = 0;
    BB2 = 1;
}
void backward(void)//后退
{
    BA1 = 0;
    BB1 = 1;
    BA2 = 1;
    BB2 = 0;
}
void stop(void)//停止
{
    BA1 = 1;
    BB1 = 1;
    BA1 = 0;
    BB1 = 0;
    BA2 = 1;
    BB2 = 1;
    BA2 = 0;
    BB2 = 0;
}
// 检测 D11 是否为低电平（黑线），采用多次采样
bit detect_black_line()
{
    int low_count = 0;
    int i = 0;
    for (i = 0; i < SAMPLE_TIMES; i++)
    {
        if (D11 == 1)
        {
            low_count++;
        }
        delay_ms(10);  // 每次采样间隔 10ms
    }
    return low_count >= THRESHOLD;
}

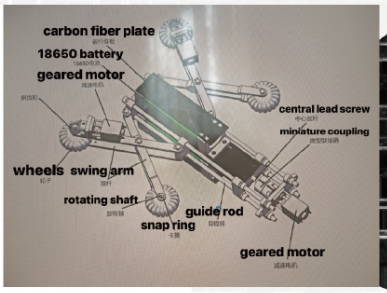
int Line_Flag = 0;
bit is_returning = 0;
//-----------------------------------------------------------------
// 函数: void main(void)
// 描述: 主函数.
// 参数: none.
// 返回: none.
// 版本: V1.0, 2015-1-12
//-----------------------------------------------------------------
void main(void)
{

    P_SW2 |= 0x80;  //扩展寄存器(XFR)访问使能
    // UART1_config(1);    // 选择波特率, 2: 使用Timer2做波特率, 其它值: 使用Timer1做波特率.

    GPIO_Init();
    // Timer0_init();
    // Timer1_init();
    EA = 1; //允许总中断

    // PrintString1("STC8H8K64U UART1 Test Programme!\r\n");  //UART1发送一个字符串

    forward();
    while (1)
    {
        if (detect_black_line())//检测到黑线
        {
            stop();
            delay_ms(1000);
            if(!is_returning)
            {
              Line_Flag++;
              delay_ms(1000);
              if(Line_Flag == 3)
              {
                backward();//触发第三条黑线返回
                is_returning = 1;
                Line_Flag = 0;
              }else
              {
                forward();
              }
            }else
            {
              backward();
            }
            // 等待传感器离开黑线，避免重复触发
            while (detect_black_line());
        }
    }
}
```



Important Components and Their Functions:

Geared Motors - These provide the necessary torque and speed to drive the wheels, enabling the robot to move smoothly through the pipe.
Carbon Fiber Plate - A lightweight yet durable material that forms the main structure of the robot.
18650 Battery - A rechargeable lithium battery that powers the robot, ensuring long-lasting operations.
Wheels - Designed for maximum grip, these wheels help the robot traverse through pipes.
Central Lead Screw - This part helps with the mechanical movement by providing precise control over motion.
Miniature Coupling - Connects the rotating shaft to the motors, ensuring efficient transfer of power.

Bill of Materials (BOM):

The BOM includes the detailed list of all parts, their quantities, and specifications. It ensures the team knows exactly what materials are required for the construction of the robot.

Brunel University London

BillOfMaterials-Team

Date	Groupprojectshouldlist ALL students' names and IDs.	Level
	Team 2 (five members) Student identifiers omitted	Year 2

Theme	Module	Project Title
Mechanical Engineering	ME2628 - Team Design Project	Team Design and Validation of

Product Code	Description	Supplier and website	Quantity	Unit Price	Total
1	Tensioning structure motor 42 s	https://item.taobao.com/item.	2	20	40
2	Stepper motor drive	https://item.taobao.com/item.	2	24.5	49
3	Stepper motor drive module	https://item.taobao.com/item.	1	2.99	2.99
4	2.0A brushed drivers	https://detail.tmall.com/item.	2	4.29	4.29
5	Development board	https://item.taobao.com/item.	1	2.17	2.17
6	The obstacle avoidance of the	https://item.taobao.com/item.	2	6.72	6.72
7	3D printed parts	https://a.m.taobao.com/i6975	14	14.28	200
8	11.1V triangular 18650 lithium	https://item.taobao.com/item.	2	60.9	121.8
9	16mm 050 micro DC gear	https://item.taobao.com/item.	2	17.46	34.92
10	Mouse switch, microswitch,	https://item.taobao.com/item.	1	0.2	0.2
11	TFTE 18650 lithium battery	https://detail.tmall.com/item.	2	15.6	31.2
12	L298N driver board driver	https://detail.tmall.com/item.	1	16.4	16.4
13	3d printer accessory +2	https://item.taobao.com/item.	1	13.2	31.2
14	TMC2209 silent drive 2209	https://item.taobao.com/item.	1	13	13
15	3D printer 42 Stepper motor	https://a.m.taobao.com/i6954	1	1.98	1.98
16	Infrared obstacle avoidance	https://a.m.taobao.com/i6954	3	3	9
17	Arduino Uno development	https://a.m.taobao.com/i6954	1	37	37
					660.86
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Design and Engineering Principles:

The design is based on the principles of efficient energy consumption, modularity, and adaptability to different pipe sizes.
The robot's structure ensures it can maintain stability even in challenging environments, such as curved pipes or pipes with debris.

By following these guidelines, the poster clearly communicates the design, functionality, and purpose of the pipe-crawling robot, offering all the necessary details for understanding the project in a visually appealing and informative way.

Portfolio copy: student identifiers removed; team authors and technical content retained.